

“water breaks” before childbirth.) The allantois, which disposes of wastes in the reptilian egg, is incorporated into the umbilical cord in mammals. There, it forms blood vessels that transport oxygen and nutrients from the placenta to the embryo and rid the embryo of carbon dioxide and nitrogenous wastes. The fourth extraembryonic membrane, the yolk sac, encloses yolk in the eggs of reptiles. In mammals it is a site of early formation of blood cells, which later migrate into the embryo proper. Thus, the extraembryonic membranes common to reptiles and mammals exhibit adaptations specific to development within a shelled egg or a uterus.

After gastrulation is complete and any extraembryonic membranes are formed, the next stage of embryonic development begins: organogenesis, the formation of organs.

## Organogenesis

During organogenesis, regions of the three embryonic germ layers develop into the rudiments of organs. Often, cells from

two or three germ layers participate in forming a single organ, with interactions between cells of different germ layers helping to specify cell fates. Adopting particular developmental fates may in turn cause cells to change shape or, in certain circumstances, migrate to another location in the body. To see how these processes contribute to organogenesis, we'll consider *neurulation*, the early steps in the formation of the brain and spinal cord in vertebrates.

### Neurulation

Neurulation begins as cells from the dorsal mesoderm form the **notochord**, a rod that extends along the dorsal side of the chordate embryo, as seen for the frog in **Figure 47.14a**. Signaling molecules secreted by these mesodermal cells and other tissues cause the ectoderm above the notochord to become the *neural plate*. Formation of the neural plate is thus an example of **induction**, a process in which a group of cells or tissues influences the development of another group through close-range interactions (see Figure 18.17b).



▼ **Figure 47.14** Neurulation in a frog embryo.

➔ **Mastering Biology** Video: Frog Development

